# Daniel Hardesty Lewis

*ML Engineer – distributed training infrastructure, cloud-scale ML systems, tabular data*

Specializing in scalable cloud infrastructure for ML training and serving. Delivered production systems across tabular, geospatial, and time-series workloads at supercomputer scale.

## Core Expertise

8 years **ML Engineering**: Design and operate ML training and inference infrastructure at scale, from data ingestion through model deployment, monitoring, and retraining

8 years **Backend & Data**: Build schemas, ETL, and production APIs (Python/FastAPI); manage data-intensive systems with Postgres, Redis, and Elasticsearch

8 years **Distributed Systems**: Scale ML workloads across distributed cloud and HPC systems using containerization and orchestration (Docker, Kubernetes, Ray), with explicit cost-performance tradeoffs

10 years **Geospatial AI**: Spatiotemporal modeling for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Collaborate across engineering, research, and product to ship; mentor engineers through design reviews, code reviews, and instruction

## Technologies

| | |
|---|---|
| Modeling | PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT |
| LLM apps | embeddings; RAG; LangChain; LangGraph; DSPy; FAISS |
| Data stack | Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; BigQuery; Redis; Elasticsearch |
| Infrastructure | Linux; Git; Docker; Kubernetes; Modal; GCS; Terraform; Triton; Ray; Dask; Airflow; CI/CD; monitoring & observability |
| Additional | CUDA; C++; TypeScript; React |

## Experience

### Professional

Apr. 2025 – Present **Co-founder**, *Homecastr*, New York, NY

Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation; owned end-to-end workflow from source data to web delivery.

- Built an NYC VAE model (12% error vs. Zillow's 8.4%); evolved to a **v11 score-based diffusion model** (inducing tokens, per-horizon loss, DDIM). Eval via **Calibration Packet**: PIT histograms, coverage, sharpness.
- Shipped a Houston diffusion-based forecasting system with fan charts; achieved as low as ∼8% annualized compounding error, balancing design choices of error vs. trajectory realism, delivered via production web UI: homecastr.com
- Refined product positioning and execution milestones via strategic feedback from a16z, MetaProp, and others.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and schema definitions for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, and FastAPI serving backed by Supabase and Redis.
- Directed development of a **conversational AI interface** simplifying global trade workflows; enabled decision-makers to instantly **retrieve BOMs and receive proactive compliance alerts**.
- Executing customer discovery with the founder of ScoutyAI and tariff economists at Columbia Business School to validate product-market fit and pricing assumptions.
- Secured market validation by presenting the prototype to KPMG, pre-dating their subsequent launch of a commercial tariff-modeling product.

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Developed the web distribution platform end-to-end including data pipeline, tiling, hosting, delivery UX to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Co-founded summitgeospatial.com to commercialize and sustain state-level terrain models developed for the $40M Texas Disaster Information System (TDIS), delivering precision decision-support for civil engineering.
- Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating Real-Time Inundation Mapping and TDIS.
- Scaled climate and flood models on supercomputers, orchestrating million-node distributed jobs across containerized HPC clusters while managing multi-million-dollar compute budgets.
- Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
- Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
- Contributed fundamental research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors.
- Served as Technical Reviewer for Frontera Doctoral Fellows (2023) and Large-Scale Computing Pathways (2022) applications, evaluating $10M+ in computing resource allocations on one of the world's largest supercomputers.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers to improve disaster resiliency and food security modeling in East Africa.
- Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
- Received research recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

## Research

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

- Math benchmarking and finetuning of Deepseek's GRPO algorithm

Jan. 2025 – Oct. 2025 **Research assistant to Dir. Financial Engineering Ali Hirsa (Industry-sponsored research)**, *Columbia University*, New York, NY

- Developed a multi-asset CVAE latent factor model for **tabular financial data** with Skew-T Mixture priors; achieved 85% $R^2$ vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

- Integrated hydrological models into the MINT Platform during a year-long competition with MITRE to provide DARPA the "Airflow for science"

## Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, and HPC.

# Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

# Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

# Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

# Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

# Languages

Spanish Native Proficiency

French Professional Proficiency

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